A POINT OF THE TARGET OF A UNIVERSALLY INJECTIVE MORPHISM OVER AN ALGEBRAICALLY CLOSED FIELD LIFTS TO A POINT OF THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The existence of a surjective flat locally of finite type cover of a nonempty algebraic stack by the spectrum of a field. Invoked inside the counted proof (source lines 2997-2998) as a cross-chapter pointer. Produces the field extension through which the reduction is carried out.
- The characterization of universal injectivity through surjectivity of the diagonal. Invoked inside the counted proof (source lines 2999-3002). Makes the relevant morphism of fibre products surjective.
- The statement that a tensor product of a field with itself over an algebraically closed base is a domain. Invoked inside the counted proof (source lines 3003-3006) as a cross-chapter pointer. Provides the generic point used to compare the two projections.
- The statement on equal integral closures of the images of a field under two morphisms. Invoked near the end of the counted proof (source lines 3013-3016) as a cross-chapter pointer. Forces the comparison of the two images to give the desired equality of fields.

Source reference. Original reference algebra-lemma-geometrically-integral-any-integral-base-change: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3007–3007. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-geometrically-integral-any-integral-base-change, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-universally-injective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2907–2907. Navigation pointer only: the counted text refers to the Stacks item lemma-universally-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-unique-point: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2997–2997. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-unique-point, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference varieties-proposition-unique-base-field: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3017–3017. Navigation pointer only: the counted text refers to the Stacks item varieties-proposition-unique-base-field, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a universally injective morphism of algebraic stacks. Let $y : \mathrm{Spec}(k) \to \mathcal{Y}$ be a morphism where k is an algebraically closed field. If y is in the image of $|\mathcal{X}| \to |\mathcal{Y}|$, then there is a morphism $x : \mathrm{Spec}(k) \to \mathcal{X}$ with $y = f \circ x$.*

Proof. We first remark this lemma is not a triviality, because the assumption that y is in the image of $|f|$ means only that we can lift y to a morphism into $\mathcal{X}$ after possibly replacing k by an extension field. To prove the lemma we may base change f by y, hence we may assume we have a nonempty algebraic stack $\mathcal{X}$ and a universally injective morphism $\mathcal{X} \to \mathrm{Spec}(k)$ and we want to find a k-valued point of $\mathcal{X}$. We may replace $\mathcal{X}$ by its reduction. We may choose a field k' and a surjective, flat, locally finite type morphism $\mathrm{Spec}(k') \to \mathcal{X}$, see Properties of Stacks, Lemma stacks-properties-lemma-unique-point. Since $\mathcal{X} \to \mathrm{Spec}(k)$ is universally injective, we find that

$$\mathrm{Spec}(k') \times_{\mathcal{X}} \mathrm{Spec}(k') \to \mathrm{Spec}(k' \otimes_k k')$$

is surjective as the base change of the surjective morphism $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathrm{Spec}(k)} \mathcal{X}$ (Lemma lemma-universally-injective). Since k is algebraically closed $k' \otimes_k k'$ is a domain (Algebra, Lemma algebra-lemma-geometrically-integral-any-integral-base-change). Let $\xi \in \mathrm{Spec}(k') \times_{\mathcal{X}} \mathrm{Spec}(k')$ be a point mapping to the generic point of $\mathrm{Spec}(k' \otimes_k k')$. Let U be the reduced induced closed subscheme structure on the connected component of $\mathrm{Spec}(k') \times_{\mathcal{X}} \mathrm{Spec}(k')$ containing ξ. Then the two projections $U \to \mathrm{Spec}(k')$ are locally of finite type, as this was true for the projections $\mathrm{Spec}(k') \times_{\mathcal{X}} \mathrm{Spec}(k') \to \mathrm{Spec}(k')$ as base changes of the morphism $\mathrm{Spec}(k') \to \mathcal{X}$. Applying Varieties, Proposition varieties-proposition-unique-base-field we find that the integral closures of the two images of k' in $\Gamma(U, \mathcal{O}_U)$ are equal. Looking in $\kappa(\xi)$ means that any element of the form $\lambda \otimes 1$ is algebraically dependent on the subfield

$$1 \otimes k' \subset (\text{fraction field of } k' \otimes_k k') \subset \kappa(\xi).$$

Since k is algebraically closed, this is only possible if $k' = k$ and the proof is complete. □